

Homework Set 4

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Due: December 15, 2025

IMPORTANT NOTE. No late submission will be allowed for this homework, as we will post the solution to this homework right after the deadline to facilitate preparation for the final examination.

SOLVE THE FOLLOWING PROBLEMS.

Problem 1 (25pts). Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, and $\mu > 0$ be given. Suppose that the set $\mathcal{S} = \{x \in \mathbb{R}^n : Ax = b, x > \mathbf{0}\}$ is non-empty. Consider the following problem:

$$\begin{aligned} v_p^* &= \inf && c^T x - \mu \sum_{j=1}^n \ln(x_j) \\ &\text{subject to} && Ax = b, \\ &&& x \geq \mathbf{0}. \end{aligned} \tag{1}$$

Let $y \in \mathbb{R}^m$ and $s \in \mathbb{R}^n$ be the Lagrangian multipliers associated with the constraints $Ax = b$ and $x \geq \mathbf{0}$, respectively.

- (a) **(10pts).** Write down the first-order optimality conditions of Problem (1) and explain why they are necessary and sufficient for optimality.
- (b) **(15pts).** Consider the following Lagrangian dual of Problem (1):

$$v_d^* = \sup_{y \in \mathbb{R}^m} \theta(y), \tag{2}$$

where

$$\theta(y) = \inf_{x \geq \mathbf{0}} \left\{ c^T x - \mu \sum_{j=1}^n \ln(x_j) + y^T (b - Ax) \right\}.$$

Show that Problem (2) is equivalent to the following problem:

$$\begin{aligned} \sup &&& b^T y + \mu \sum_{j=1}^n \ln(c_j - (A^T y)_j) \\ &\text{subject to} && A^T y \leq c. \end{aligned}$$

Problem 2 (25pts). Let $Q \in \mathcal{S}^n$, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $c \in \mathbb{R}^n$ be given. Consider the following problem:

$$\begin{aligned} \inf &&& f(x) = \frac{1}{2} x^T Q x + c^T x \\ &\text{subject to} && Ax \leq b. \end{aligned} \tag{3}$$

- (a) **(10pts).** Let $v \in \mathbb{R}^m$ be the multiplier associated with the constraint $Ax \leq b$. Write down the KKT conditions associated with Problem (3) and explain why they are necessary for optimality.

- (b) **(15pts)**. Let $\bar{x} \in \mathbb{R}^n$ be a KKT point of Problem (3); i.e., there exists a multiplier $\bar{v} \in \mathbb{R}^m$ such that $(\bar{x}, \bar{v})$ satisfies the KKT conditions found in (a). Let $\mathcal{I}(\bar{x}) = \{i : a_i^T \bar{x} = b_i\}$, where a_i^T is the i -th row of A , be the active index set associated with $\bar{x}$. Suppose that whenever $d \in \mathbb{R}^n$ satisfies $a_i^T d \leq 0$ for all $i \in \mathcal{I}(\bar{x})$, we have $d^T Q d \geq 0$. Show that $\bar{x}$ is a local minimum of Problem (3). (*Hint: Begin by showing that $f(x) - f(\bar{x}) = \nabla f(\bar{x})^T (x - \bar{x}) + \frac{1}{2}(x - \bar{x})^T Q (x - \bar{x})$ for any $x \in \mathbb{R}^n$. Then, proceed by letting $x \in \mathbb{R}^n$ be a feasible solution to Problem (3) that lies in a sufficiently small neighborhood of $\bar{x}$.)*)

Problem 3 (20pts). Observe that Problem (37) in Example 4 of Handout 7 can be equivalently formulated as

$$\begin{aligned} v_p^* &= \text{minimize} && -x \\ &\text{subject to} && x \leq 1, \\ &&& x(x - 2) = 0. \end{aligned} \tag{4}$$

Consider the following dual of Problem (4):

$$v_d^* = \sup_{v \geq 0, w \in \mathbb{R}} \theta(v, w).$$

Here, we have

$$\theta(v, w) = \inf_{x \in \mathbb{R}} \{-x + v(x - 1) + wx(x - 2)\}.$$

- (a) **(10pts)**. Find an explicit expression for $\theta(v, w)$.
- (b) **(10pts)**. Using the result in (a), determine the value v_d^* . Show your derivation.

Problem 4 (30pts). Let $A \in \mathcal{S}^n$ be given. Consider the following QCQP:

$$\begin{aligned} &\text{minimize} && x^T A x \\ &\text{subject to} && x_i^2 = 1 \quad \text{for } i = 1, \dots, n. \end{aligned} \tag{5}$$

- (a) **(5pts)**. Derive the semidefinite relaxation of Problem (5) using the techniques introduced in class.
- (b) **(10pts)**. Write down the dual of the semidefinite relaxation you found in (a). Does the primal-dual pair of SDPs you obtained have zero duality gap? Justify your answer.
- (c) **(15pts)**. The Lagrangian dual of Problem (5) is given by

$$\sup_{w \in \mathbb{R}^n} \theta(w), \tag{6}$$

where

$$\theta(w) = \inf_{x \in \mathbb{R}^n} \left\{ x^T A x + \sum_{i=1}^n w_i (1 - x_i^2) \right\}.$$

Find an explicit expression for $\theta(w)$. Hence, show that Problem (6) is equivalent to the semidefinite relaxation dual you found in (b).